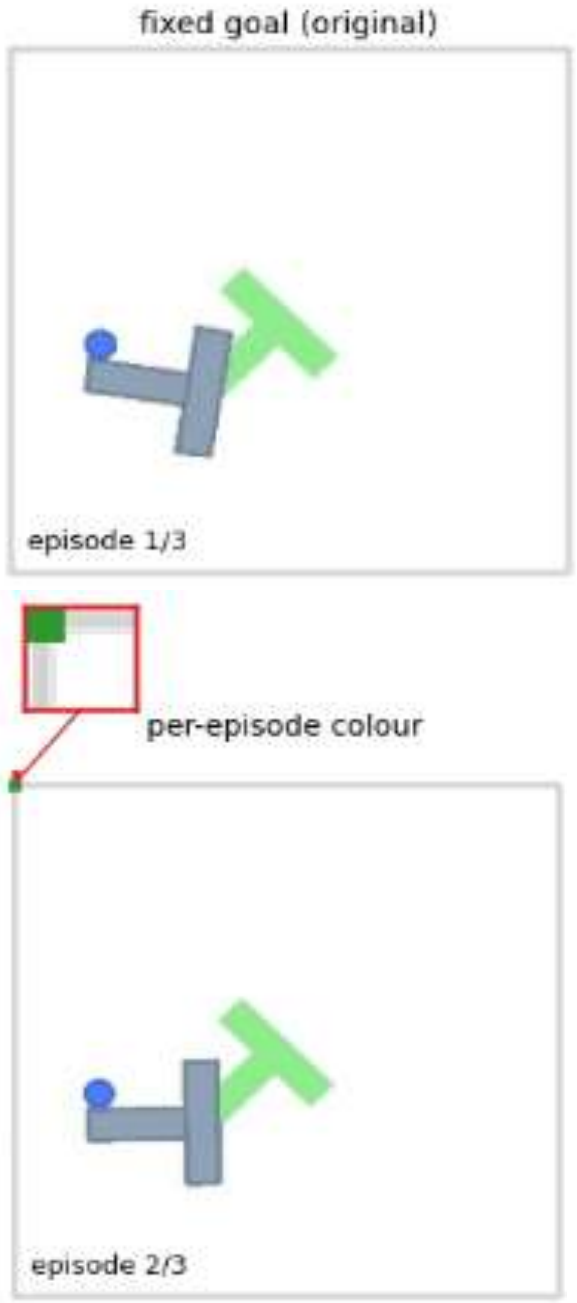
How Do We Fix an Obsessed Encoder?

Protecting Control-Relevant Geometry from Predictive Shortcuts

Changchen Song
Jiahang Cao



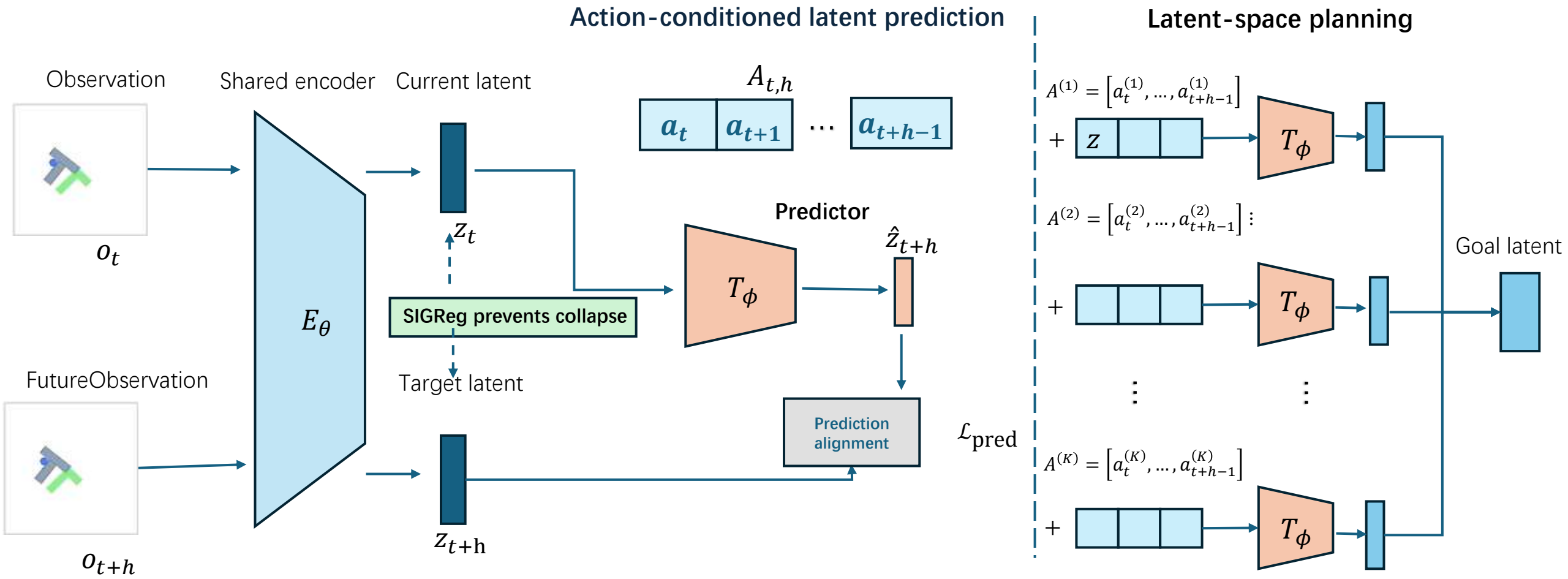
Predictive learning needs a control criterion

Abstract

Predictive alignment naturally favors features that remain stable across paired observations. In control data, however, the most persistent signal may be an episode-constant nuisance rather than the physical state affected by actions.

A non-collapsed representation can therefore preserve what is easy to predict while allocating its geometry to the wrong factor, scrambling what is controllable and undermining latent-space planning.
We use action-derived control evidence to preserve control-relevant geometry, enabling robust planning without abandoning predictive learning.

From JEPA to a latent world model



$$\mathcal{L}_{\text{pred}} = \ell(\hat{z}_{t+h}, z_{t+\text{h}})$$

$$A^* = \arg\min_A \; d\big(\hat{z}_{t+h}(A), z_g\big)$$

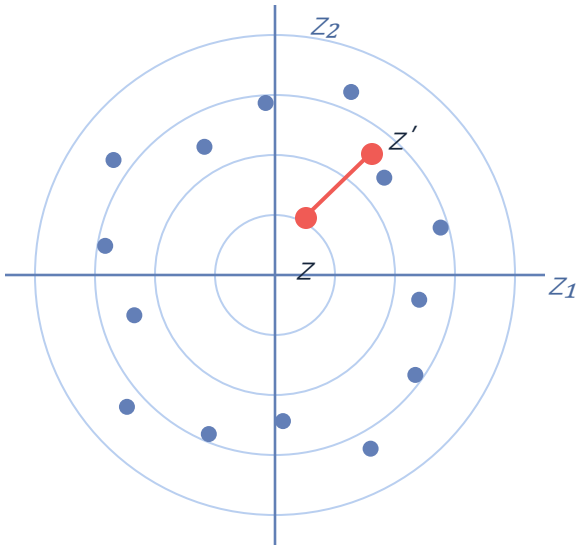
A useful world model must predict future latent states while preserving a geometry suitable for planning.

Identifiable world geometry under ideal conditions

Under the theorem assumptions

World latent

z in R²

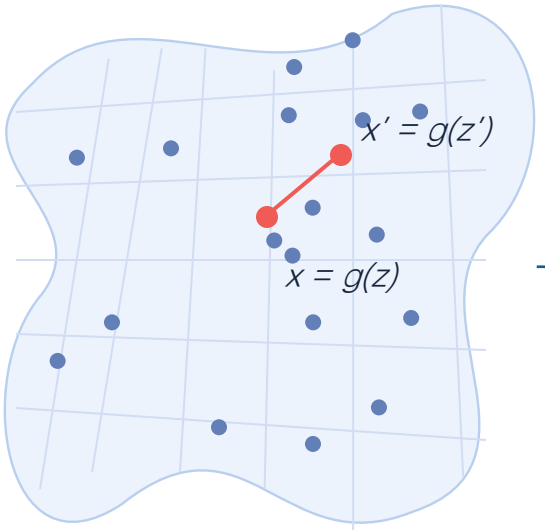


unknown nonlinear observation process

$$x = g(z)$$

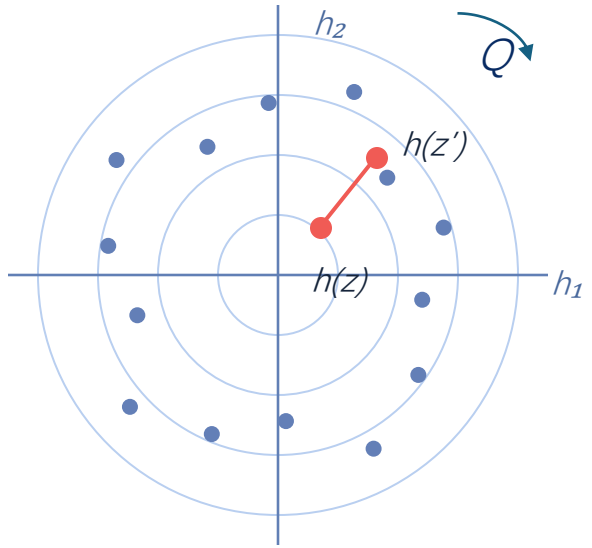
Observation

x in R²



Learned representation

h(z) in R²



Assumptions

1. Prediction alignment

$$\mathcal{L}_{\text{align}} = \mathbb{E}\,\|h(z') - h(z)\|^2$$

2. Gaussian regularization

$$h(z) \sim \mathcal{N}(0, I)$$

Theorem (identifiability)

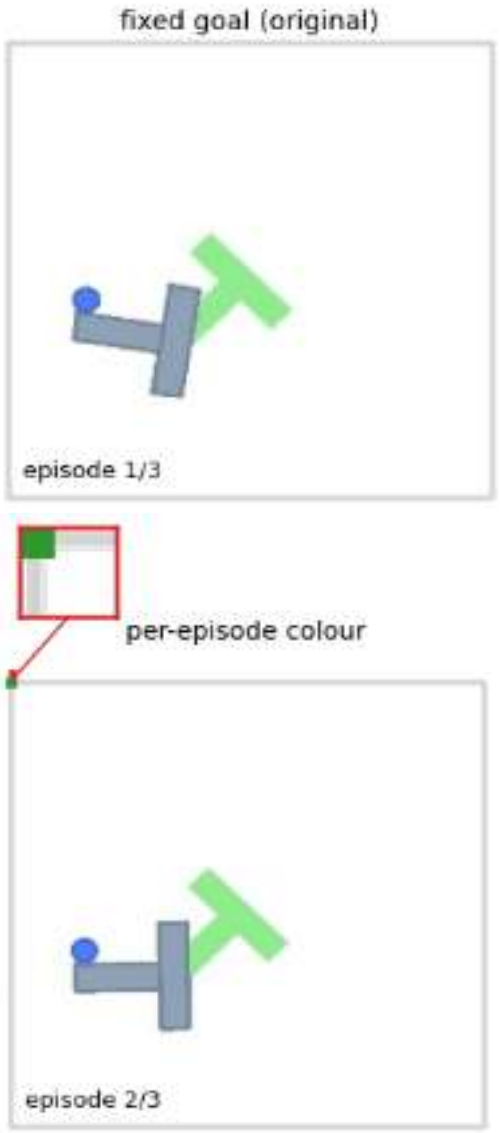
Under the above assumptions

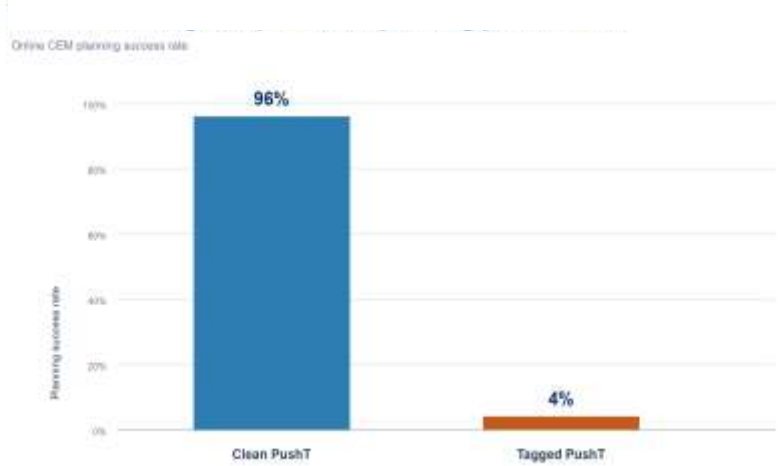
$$h(z) = Qz, \quad Q^\top Q = I$$
$$\|h(z_i) - h(z_j)\| = \|z_i - z_j\|$$

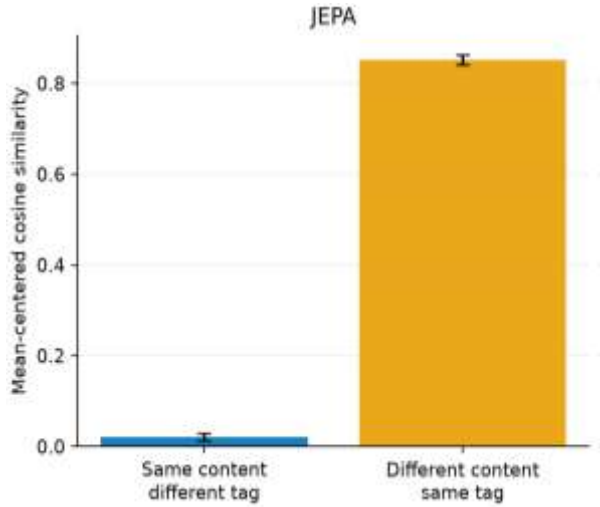
Orthogonal identifiability makes Euclidean latent distance meaningful for planning.

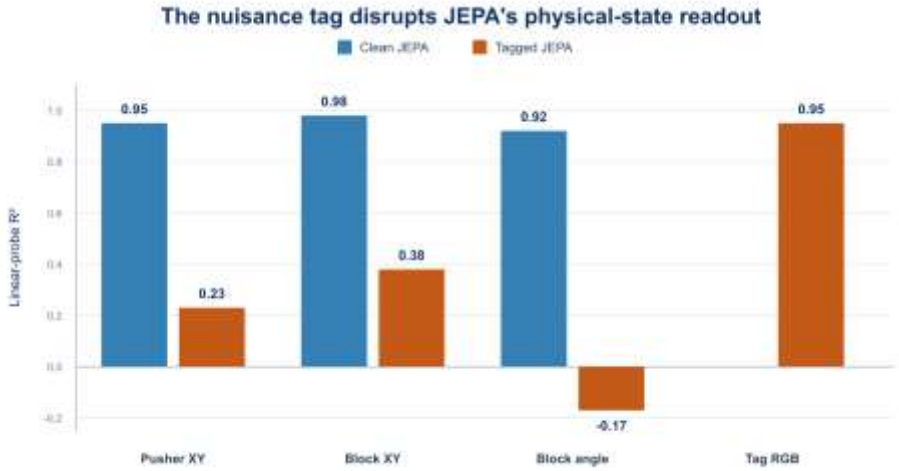
The nuisance tag dominates JEPA's learned representation

Tagged PushT changes only an episode-constant 5×5 RGB patch



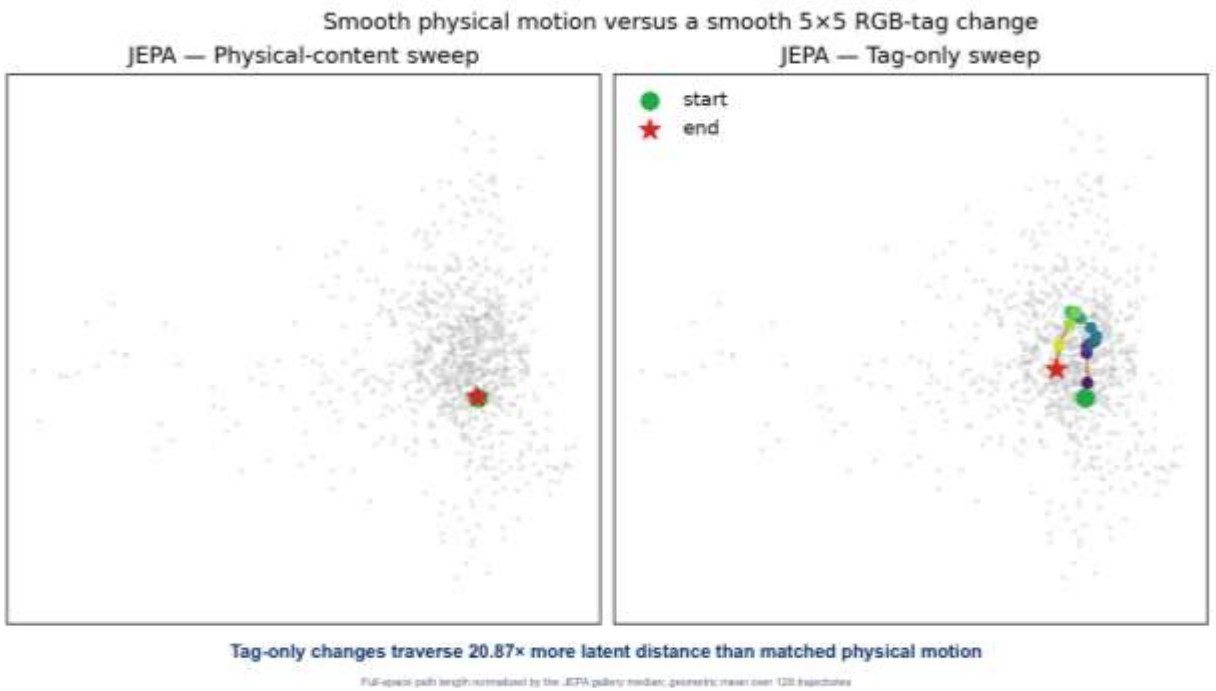






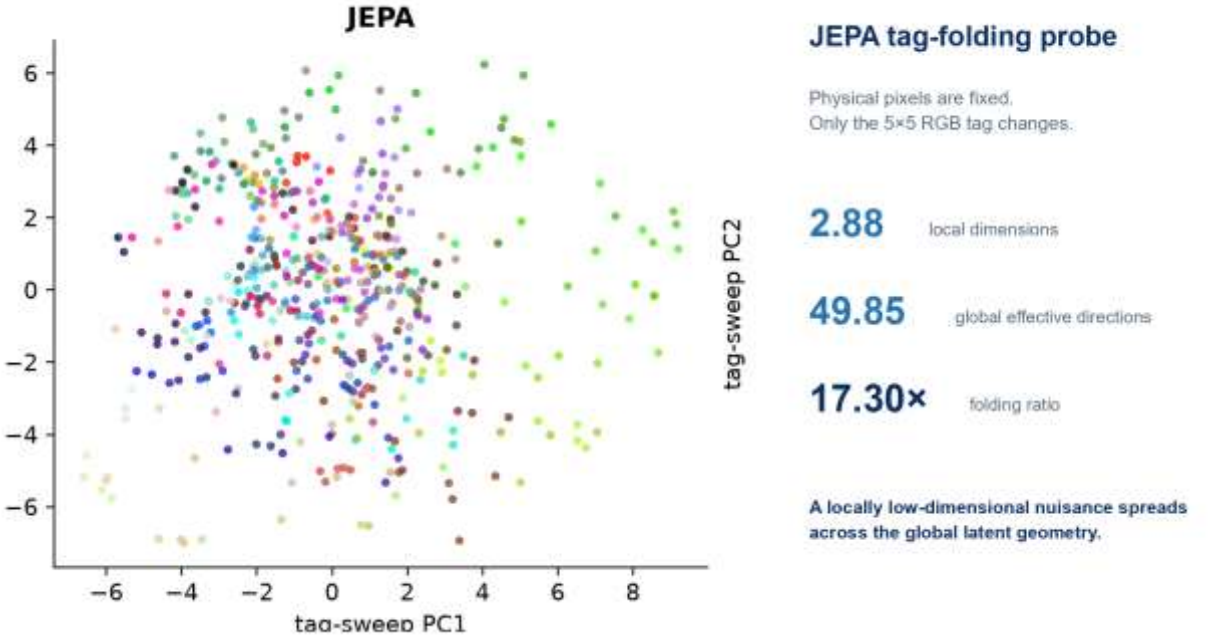
Despite a well-optimized prediction loss, three diagnostics reveal that the nuisance tag dominates JEPA's representation:

- **Shared-Content similarity: 0.021** ↑
- **Shared-Tag similarity: 0.852** ↓
- **Planning success rate: 4%** ↑ *(96% on Clean PushT)*



Latent displacement allocation

JEPA is nearly insensitive to physical motion, yet changes substantially with the nuisance tag: tag variation induces **20.87× greater latent displacement** than matched physical motion.



Tag-folding phenomenon

Although tag variation is locally only **2.88-dimensional**, it is folded across **49.85 global effective directions**. SIGReg constrains the aggregate latent distribution—not which factor occupies that capacity—so the representation can remain non-collapsed and approximately Gaussian while its geometry is dominated by the nuisance tag.

SIGReg prevents statistical collapse, but not semantic misallocation.

Predictive Learning Favors Slow Features—even When They Are Nuisances

Idealized guarantee: under isotropic Gaussian latents with a common temporal correlation ρ, alignment with SIGReg recovers the world latent up to rotation.

Physical state



changes with the dynamics

$$\rho(\text{physical}) < 1$$

Episode tag



unchanged within the episode

$$\rho(\text{tag}) \approx 1$$

Tagged PushT breaks the shared-correlation condition. The physical state evolves, but the episode tag remains nearly constant.

For a standardized scalar feature u:

$$L_{\text{align}}(u) = \mathbb{E}[(u_t - u_{t+h})^2] = 2\big(1 - \rho(u)\big)$$

$$\rho(\text{tag}) > \rho(\text{physical}) \Longrightarrow L_{\text{align}}(\text{tag}) < L_{\text{align}}(\text{physical})$$

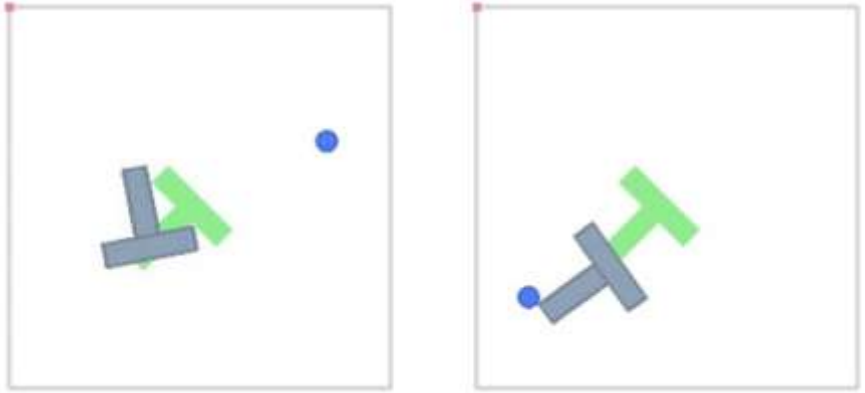
A persistent nuisance is cheaper to align than the control-relevant state

Theory: Klindt, LeCun & Balestriero (2026), When Does LeJEPA Learn a World Model?

Prediction alone does not identify control relevance

What prediction rewards

$o_t = g(x_t, c_e)$ $\quad$ $c_{t+h} = c_t$



t $\quad$ $t + h$

physical state x_t	changes with dynamics	$\rho_x < 1$
episode tag c_e	Persists unchanged	$\rho_c \approx 1$

Prediction favors the more persistent signal.

Predictability
≠
Control relevance

A non-collapsed representation can still allocate capacity to the wrong factor.

SIGReg prevents trivial collapse but does not choose what the representation should preserve.

What control evidence identifies

Inverse action

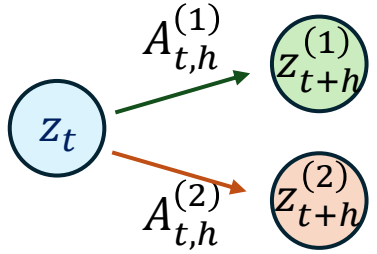
$I_\psi(z_t, z_{t+h}) \rightarrow \hat{A}_{t,h}$



Which action connects the states?

Reachability

$R_\omega(z_t, A_{t,h}, Z) \rightarrow s_{t,h}$



Which future state can the action reach?

$$L_c = L_{\text{inv}} + \lambda_r L_{\text{reach}}$$

Actions reveal latent changes that support interaction.

Control evidence supplies the selection criterion missing from prediction alone.

It determines which predictive updates may modify the shared representation.

Control evidence defines what the representation must preserve

Predictive evidence

$$(z_t, A_{t,h}) \longrightarrow T_\phi \longrightarrow \hat{z}_{t+h} \qquad [L_{\text{pred}}]$$

What can be predicted?

Action-derived control evidence

Inverse action $(z_t, z_{t+h}) \longrightarrow \hat{A}$

Reachability $(z_t, A_{t,h}) \longrightarrow$ future−state identity

$$L_c = L_{\text{inv}} + \lambda_r L_{\text{reach}}$$

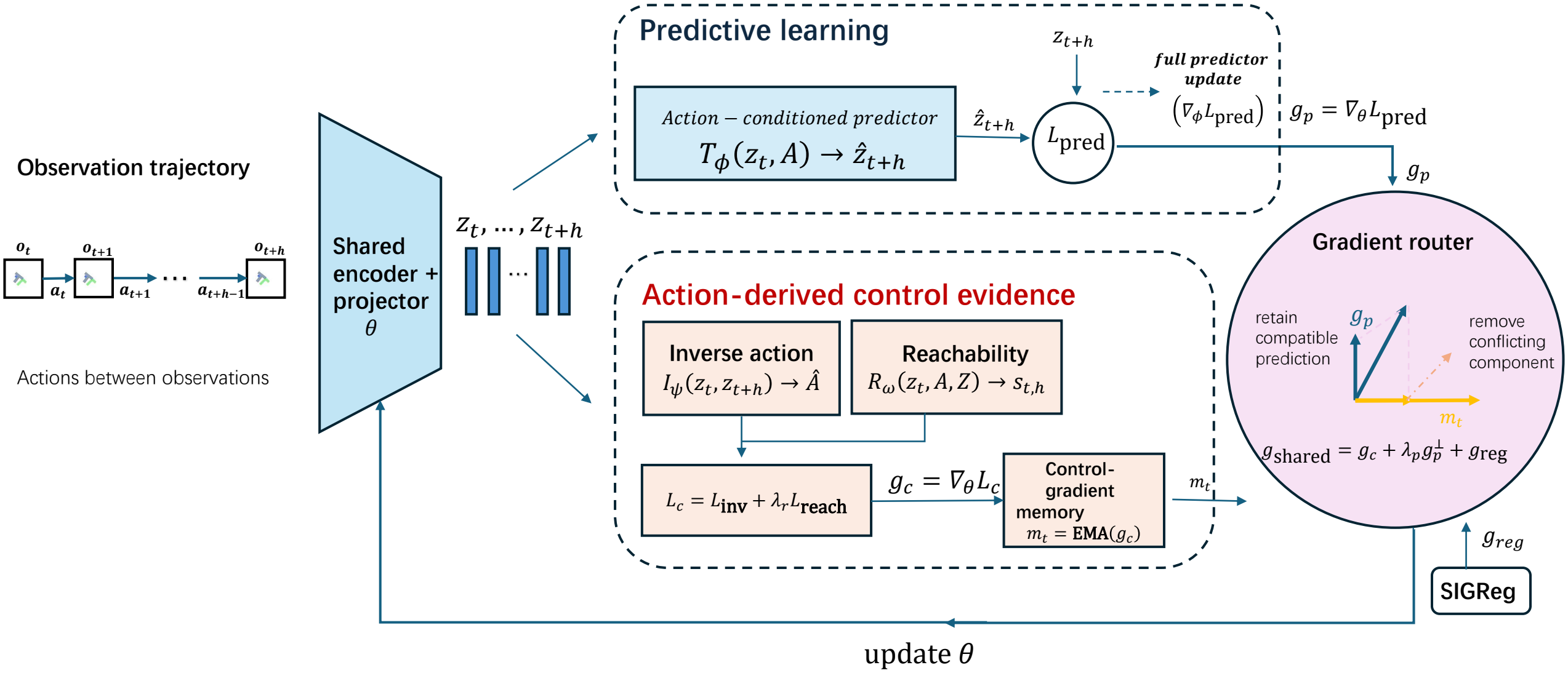
What must remain controllable?

prediction proposal —control−guided routing, preserve control geometry→ *compatible* update

Control evidence filters how prediction may update the shared encoder.

Prediction supplies useful variation. Action evidence supplies the preservation criterion.

Control-supported representation learning



Control evidence determines which directions predictive learning may change.

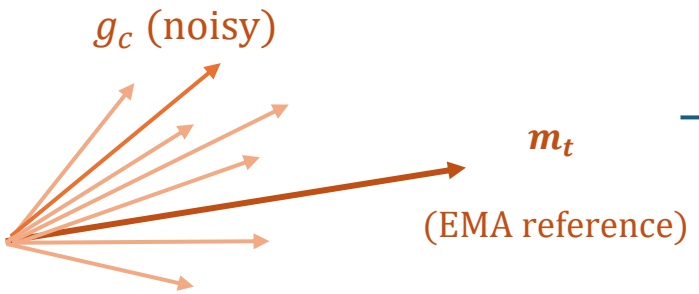
Mathematical guarantee: local first-order protection

The routed prediction update is locally neutral to the control objective.

1. Control reference

$$g_c = \nabla_\theta L_c$$

$$m_t = \beta m_{t-1} + (1-\beta) g_c$$

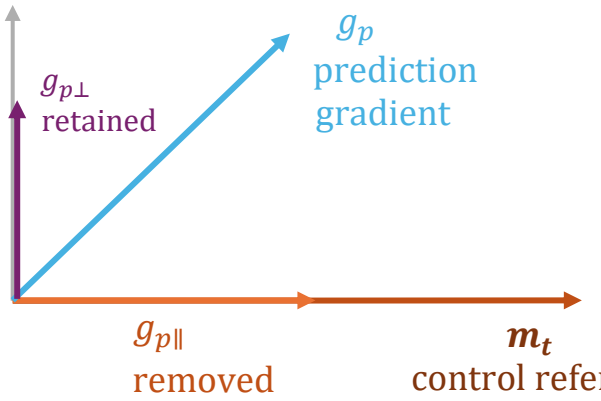


EMA provides a stable local reference in parameter space.

m_t approximates g_c

(parameter space θ)

2. Route the prediction gradient

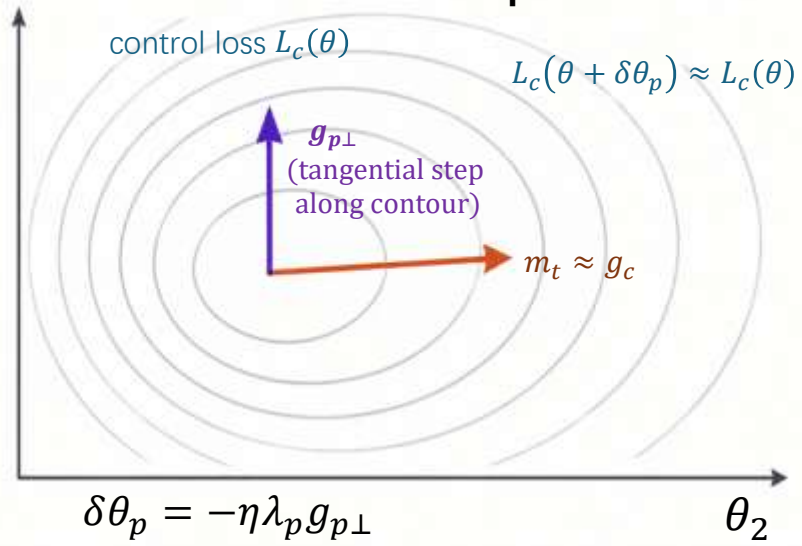


$$g_{p\perp} = g_p - \frac{\langle g_p, m_t \rangle}{\|m_t\|^2 + \varepsilon} m_t$$

$$\langle m_t, g_{p\perp} \rangle \approx 0$$

(parameter space θ)

3. First-order consequence



$$\Delta L_c|_{\text{pred}} = \nabla_\theta L_c^\top \delta\theta_p + O(\eta^2)$$

$$\approx -\eta\lambda_p \langle m_t, g_{p\perp} \rangle + O(\eta^2)$$

$$\mathbf{\Delta L_c|_{pred} = O(\eta^2)}$$

(parameter space θ)

Shared encoder

$$g_{\text{shared}} = g_c + \lambda_p g_{p\perp} + g_{\text{reg}}$$

Prediction remains fully trainable in the predictor and in parameter directions orthogonal to the control reference.

Predictor-private parameters

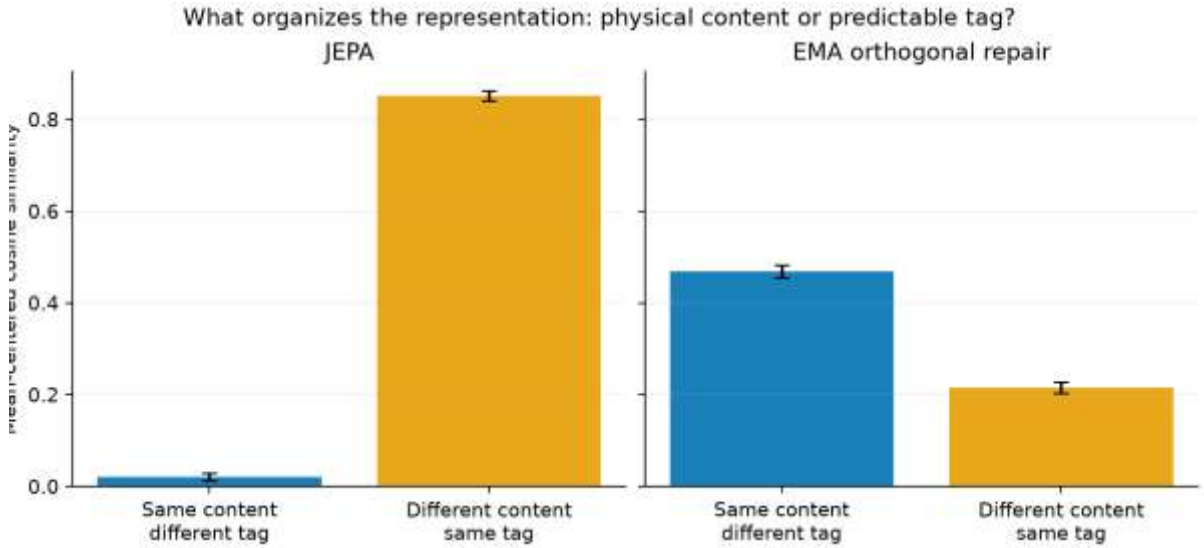
$$g_\phi = \nabla_\phi L_{\text{pred}}$$

Control evidence protects one local parameter direction without freezing predictive learning.

Tagged PushT: Planning and Representation Results

Frozen representations · Online CEM planning · Matched tag interventions



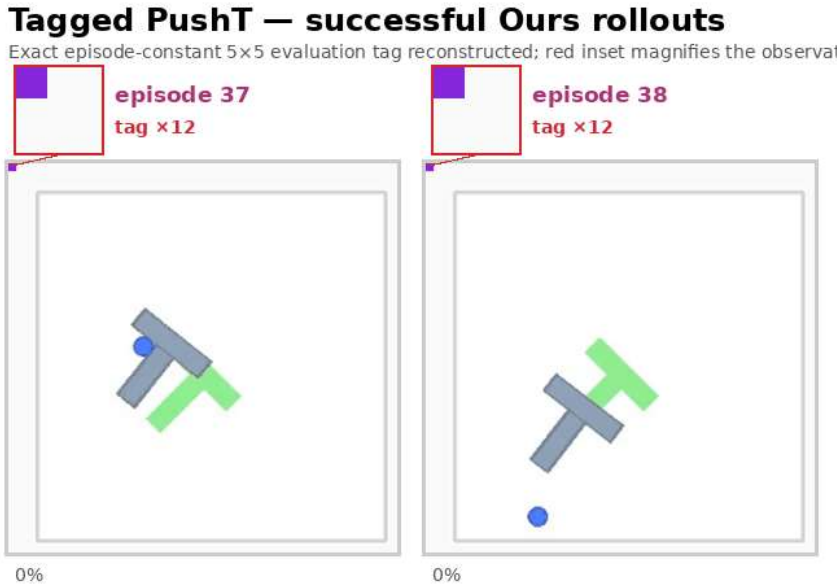


Ours is the only method that simultaneously achieves high planning success (88%) and reverses tag dominance in the latent representation (0.469 > 0.215).

Tagged PushT: successful planning under a persistent nuisance



Same start and goal: JEPA fails, Ours succeeds

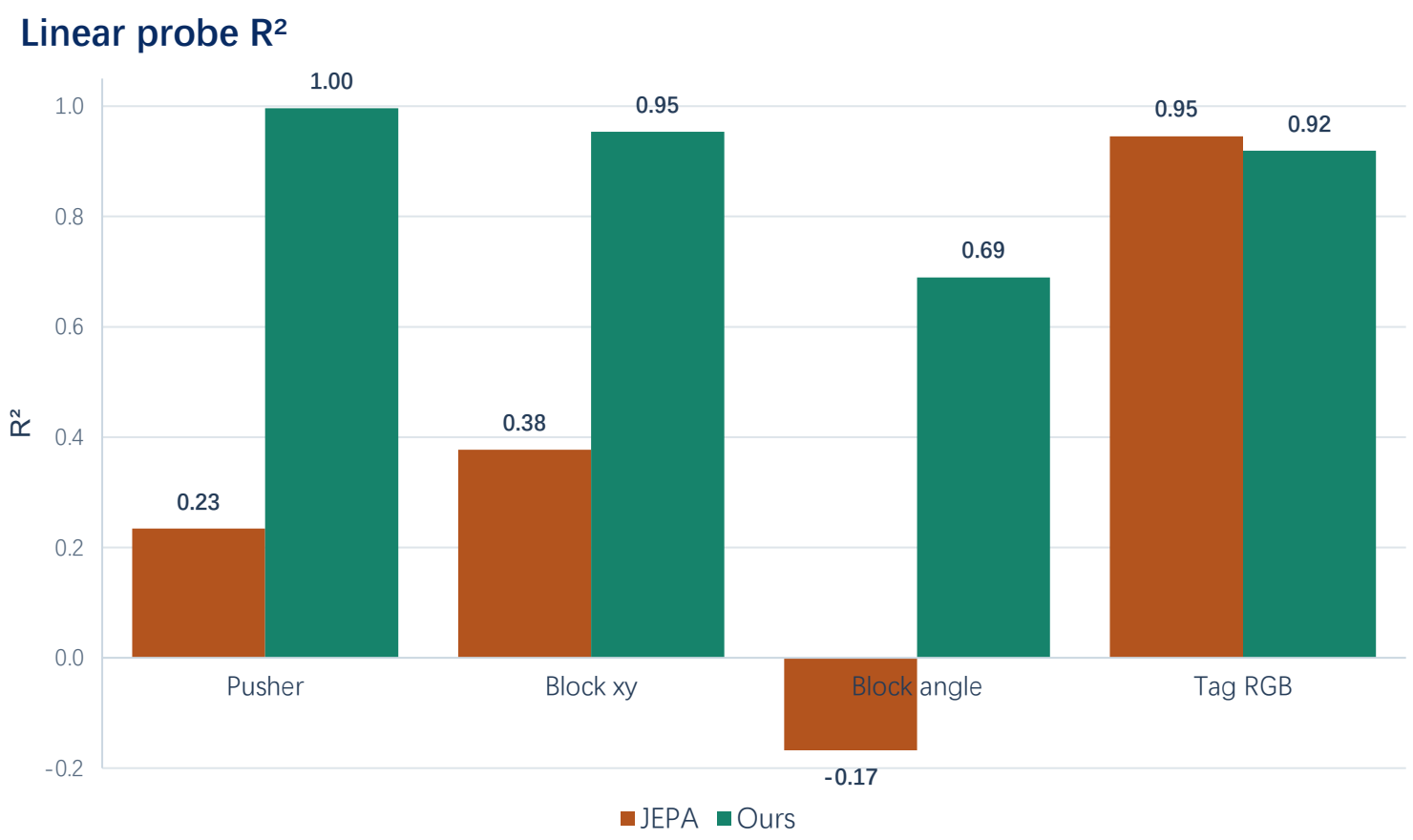


Representative successful Ours rollouts

Under the episode-constant tag, Ours completes the manipulation in matched cases where JEPA fails.

Ours restores physical state without erasing the tag

Frozen projection-space linear probes on tagged PushT



Physical state becomes accessible

0.15
JEPA physical mean R^2

0.88
Ours physical mean R^2

The tag remains readable

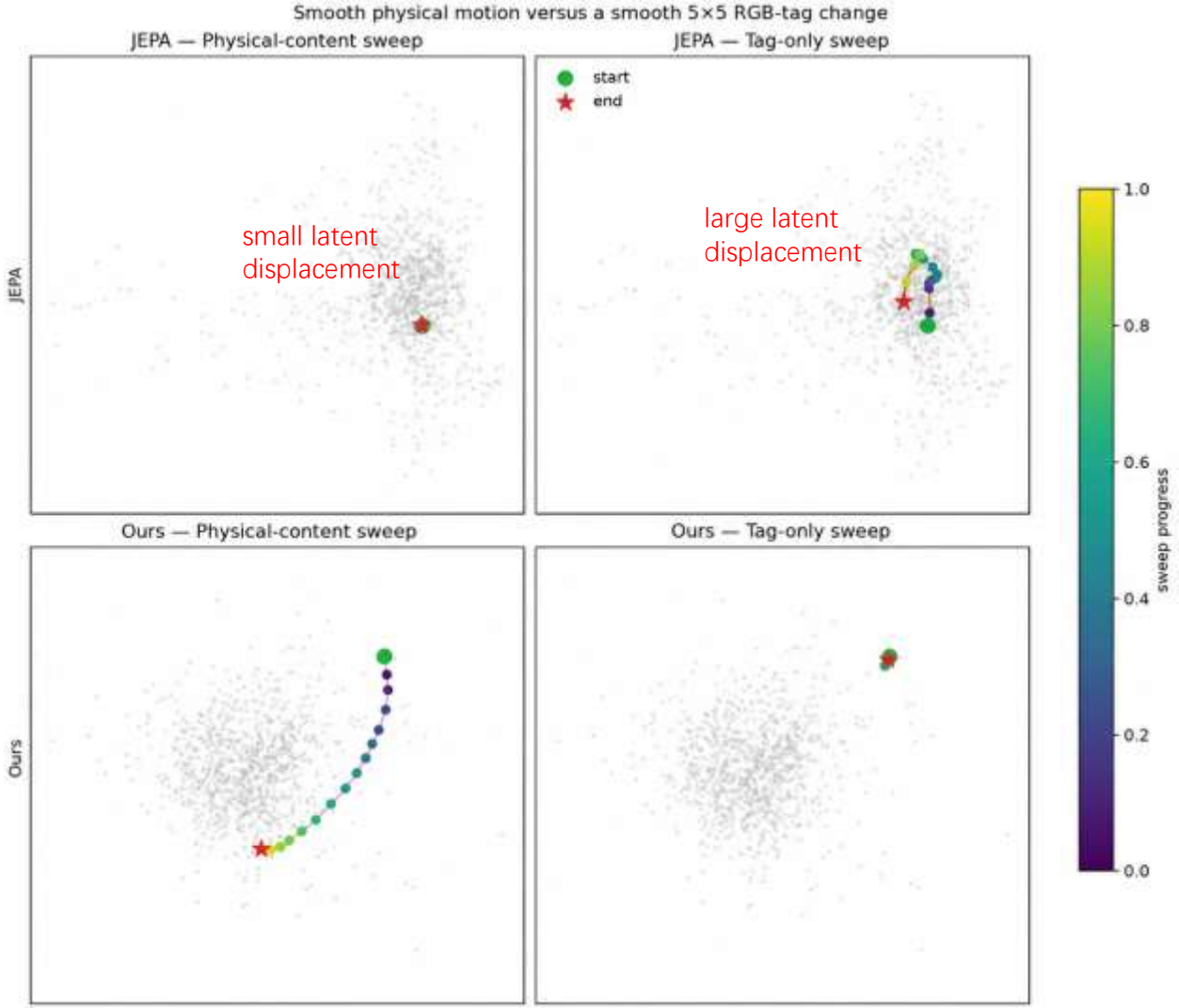
0.95
JEPA tag R^2

0.92
Ours tag R^2

The improvement cannot be explained by simply filtering out the nuisance tag.

Probe split seeds 17, 73, 137; one training seed

Control evidence changes which information dominates the representation, not merely which information is present.



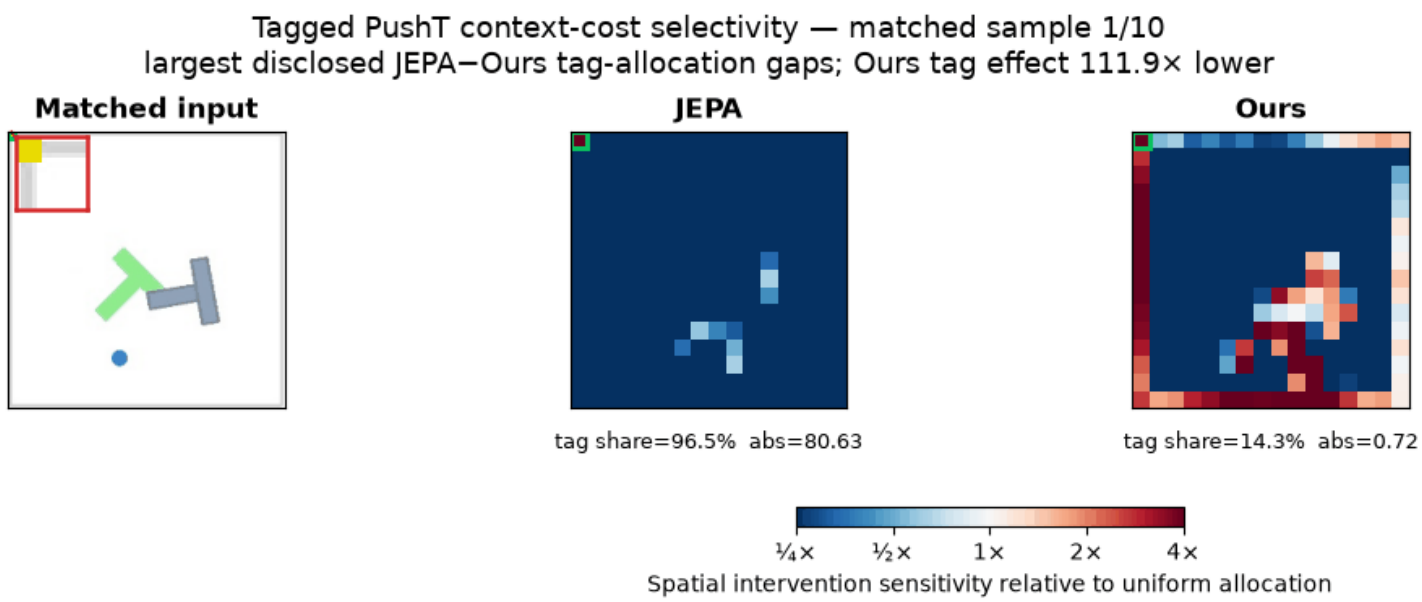
Representative 2-D PCA visualization

JEPA: In this 2-D projection, physical motion produces little visible displacement, while tag variation traces a more pronounced trajectory.

Ours: Physical motion traces a clear trajectory, while tag-induced changes appear more localized.

The visualization suggests a shift toward physical-state sensitivity.

Ours redirects planning sensitivity away from the nuisance tag



Controlled intervention

We perturb one spatial tile at a time and measure the resulting change in candidate planning costs.

JEPA

The nuisance tag receives 91.8% of the spatial sensitivity, with an absolute tag effect of 68.43.

Ours

Tag share falls to 17.6%, while the absolute tag effect drops to 0.77. Sensitivity shifts toward the task-relevant scene content.

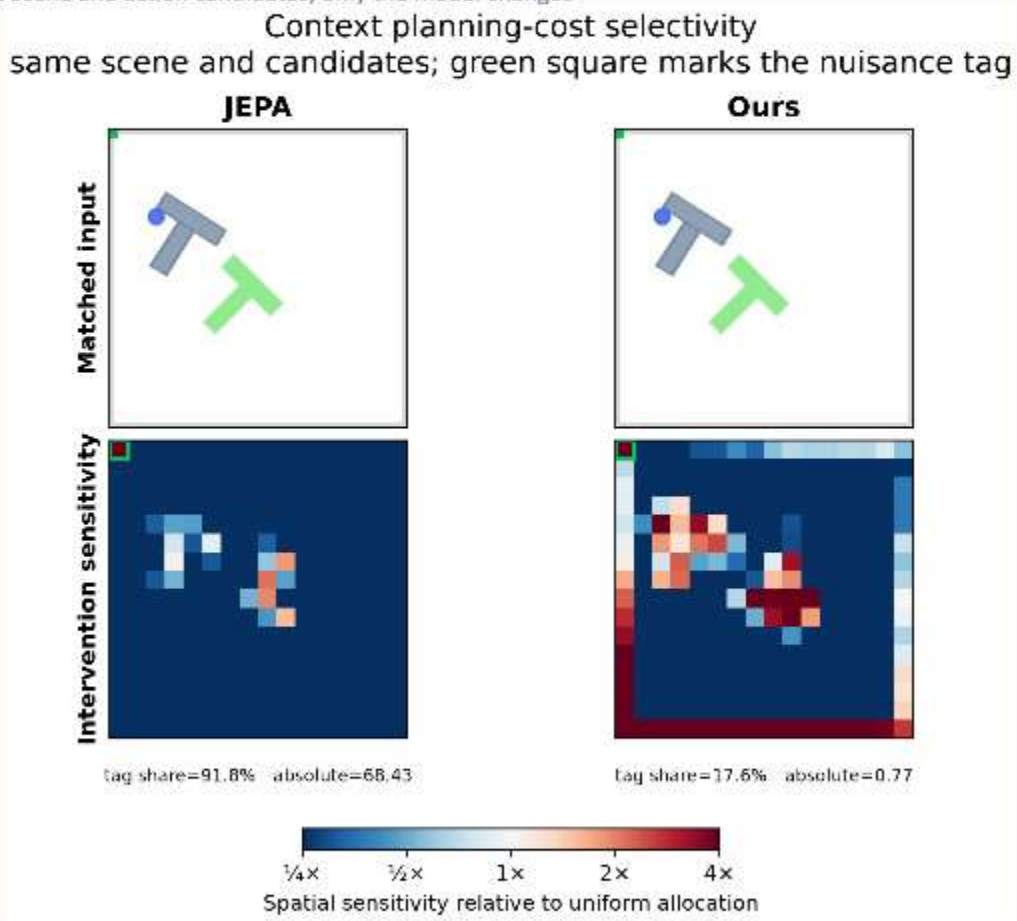
The tag remains visible, but it no longer dominates the planner's cost function.

The method changes what the representation is sensitive to

Matched interventions reveal less nuisance allocation and more physical selectivity

Spatial planning-cost selectivity

Same scene and action candidates; only the model changes



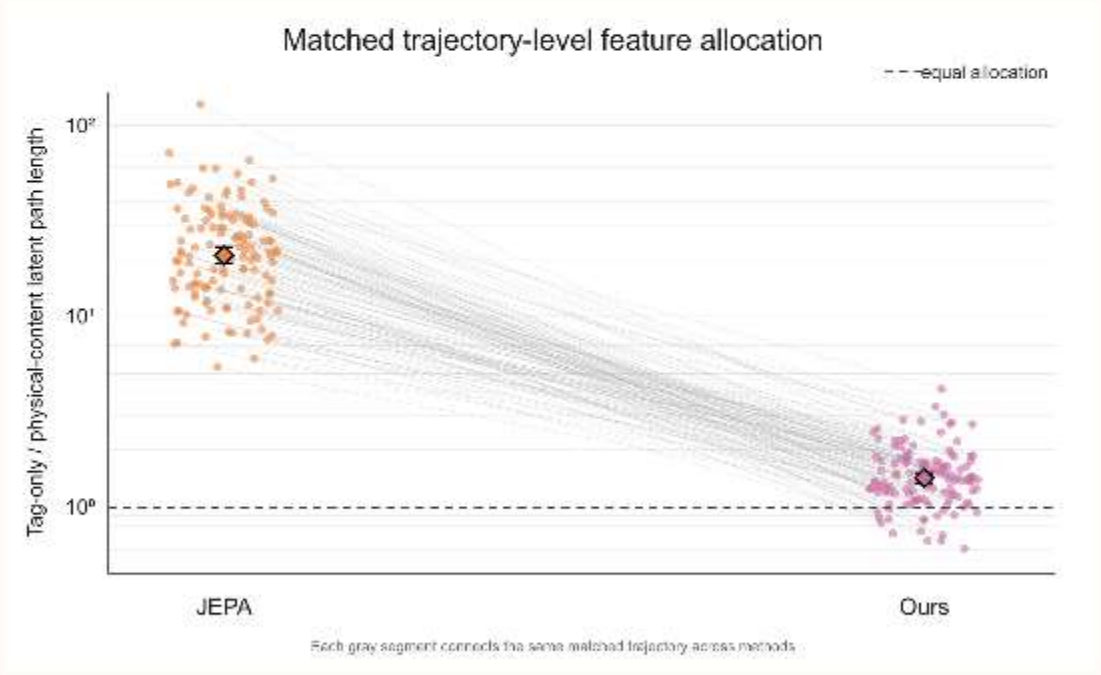
Matched trajectory-level allocation

Absolute tag effect

68.43 → 0.77

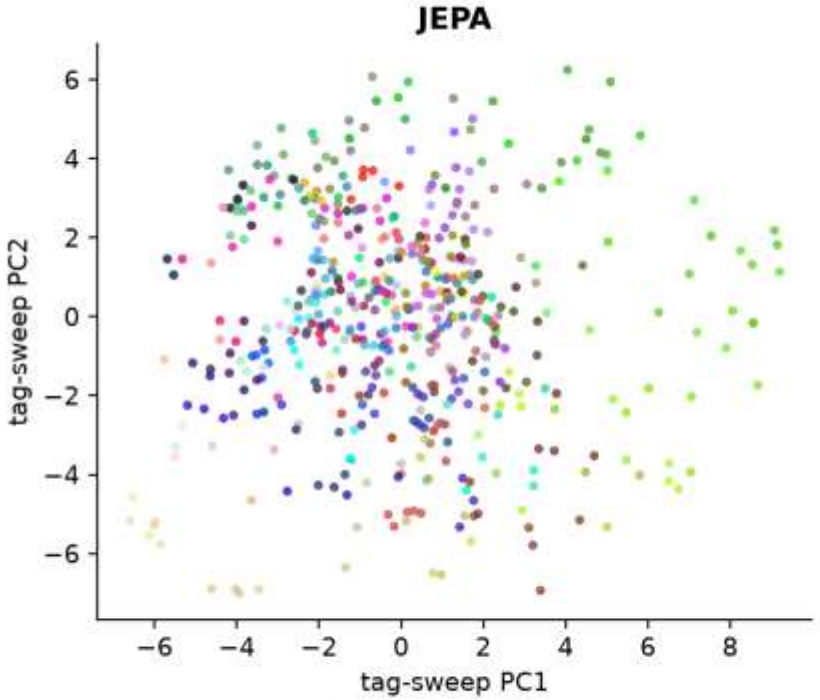
Tag / physical path ratio

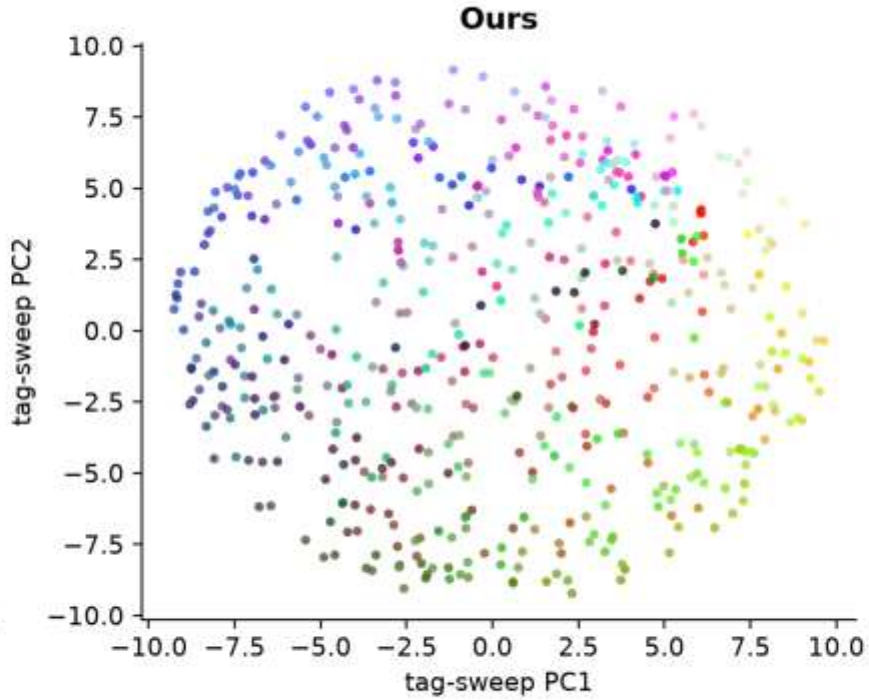
20.87 → 1.42



Ours does not merely improve task success. It reduces nuisance dominance in the learned geometry.

Does a low-dimensional tag spread across the latent space?



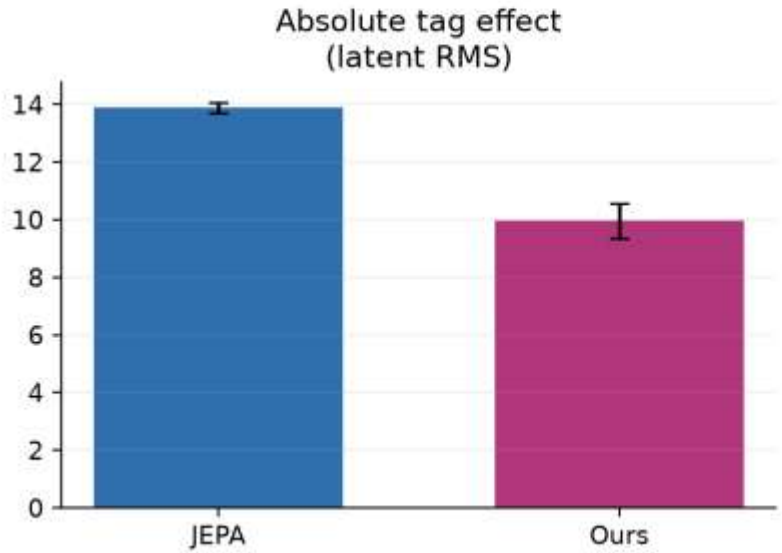


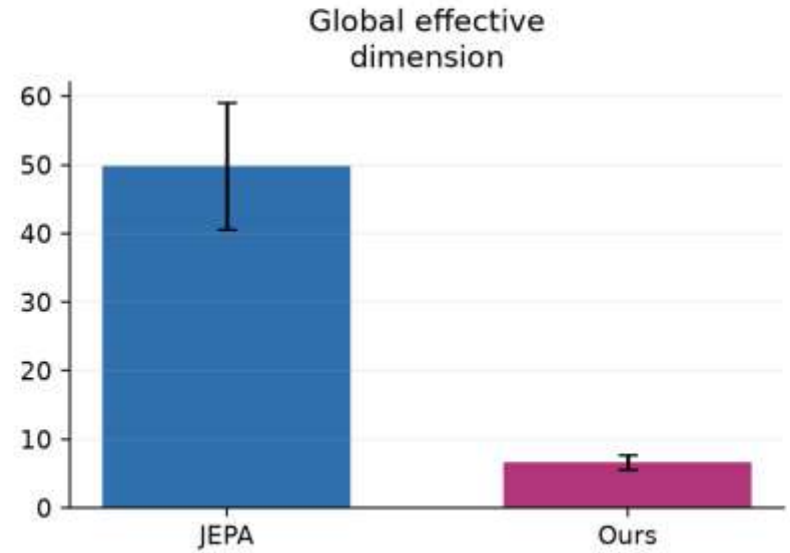
Controlled intervention

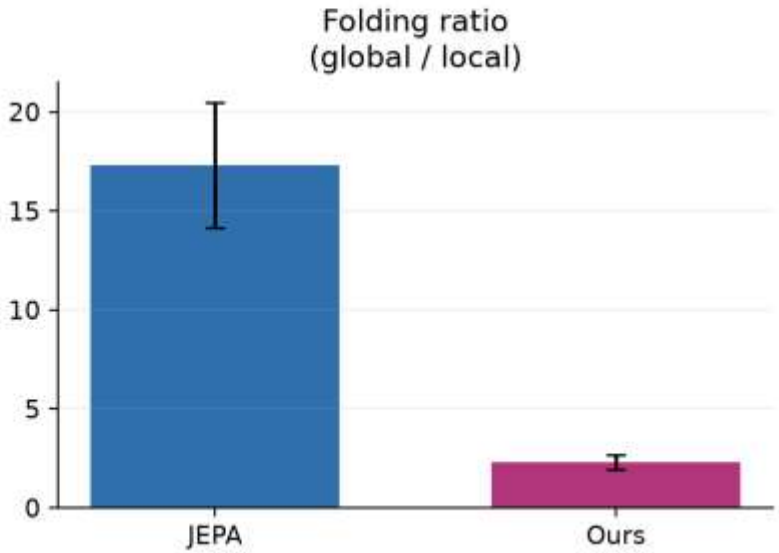
Physical pixels fixed
Only the 5×5 RGB tag changes
No retraining or simulator labels

Folding signature

large global PCA dimension
÷ small local TwoNN dimension



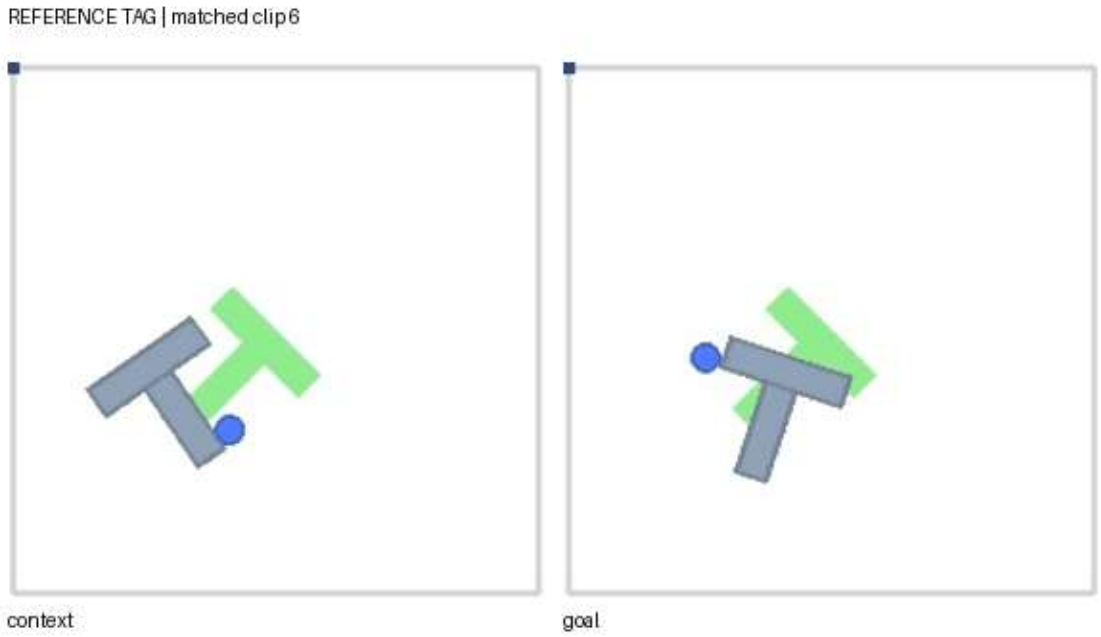




Ours reduces tag-induced selection changes by about 10×

Physical scene, goal, and 32 candidate action sequences remain fixed

Matched counterfactual intervention



Controlled change

Only the 5×5 RGB tag changes

Planner response

Selected candidate changed

JEPA	Ours
69.53%	7.03%

Normalized cost distortion

JEPA	Ours
58.36%	10.78%

Pair-order reversal

JEPA	Ours
9.84%	2.27%

The nuisance tag remains visible, but it no longer controls the planner's ranking and selection.

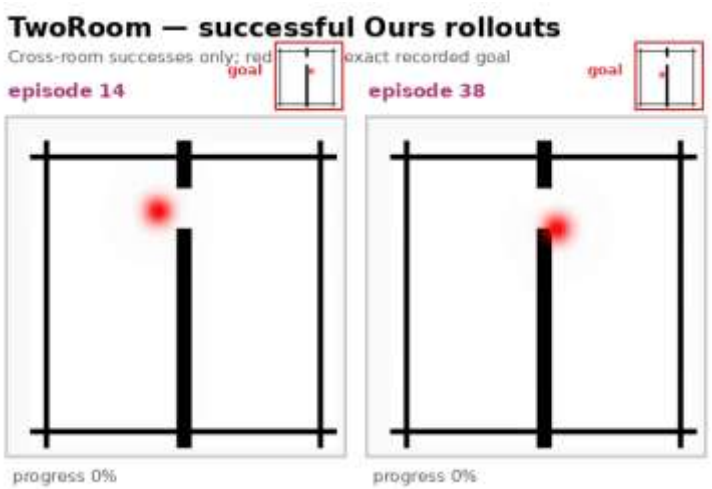
Control-supported learning preserves strong planning capability

Ours exceeds JEPA on three of four clean tasks while remaining competitive on Clean PushT.



Successful behavior across control tasks

Representative closed-loop CEM rollouts using the frozen representation



TwoRoom 96%

48 / 50 episodes



Reacher 88%

44 / 50 episodes



Cube 80%

40 / 50 episodes

Pilot protocol: epoch-10 checkpoint, training seed 0, evaluation seed 42

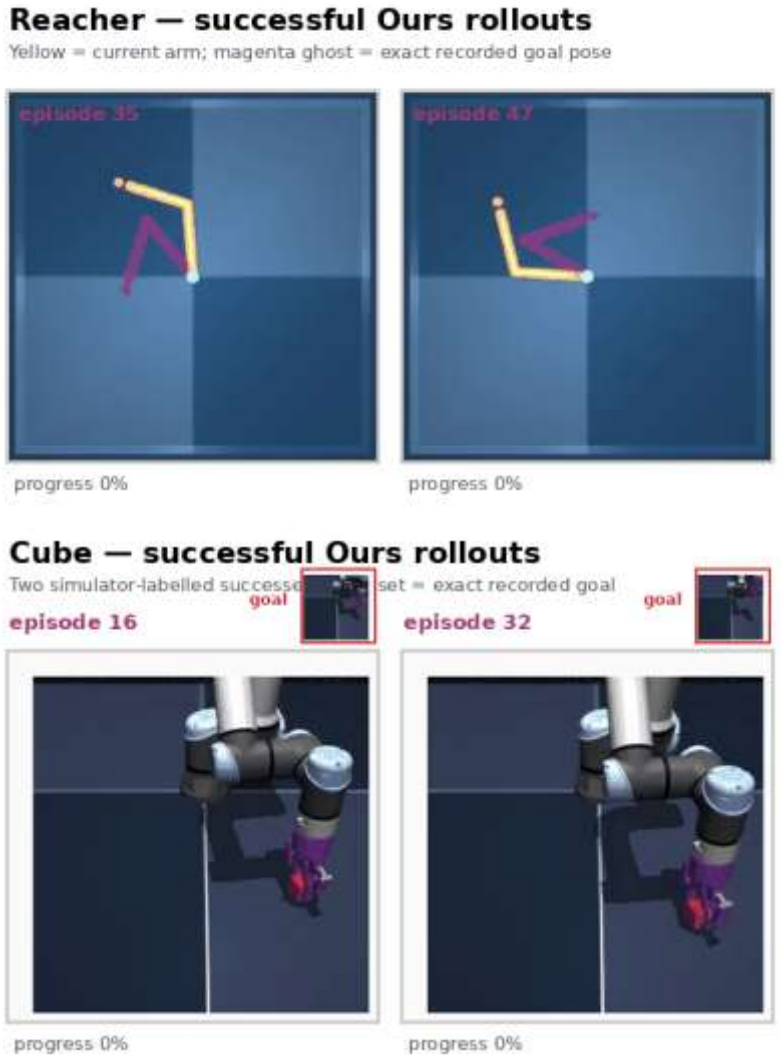
GIFs contain two selected simulator-labelled successes per task

The same control-supported representation enables reliable closed-loop behavior beyond tagged PushT.

From readable features to decision-useful representations

Action-derived control evidence limits the dominance of predictive shortcuts in the shared representation.

Readable: Physical state remains recoverable from the frozen representation.
Dominant: Shared physical content, rather than tag identity, organizes the learned geometry.
Selectively used: Counterfactual tag changes exert substantially less influence on planning, while task performance is retained.
Useful: The resulting representation supports robust planning on Tagged PushT and remains competitive on clean tasks.



Predict what you can.
Represent what changes decisions.